

Seanie Stack

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Professional Summary

Final Year BSc Computer Systems student with 8 months of professional software development experience with a focus on backend and systems engineering. Designed and build a full programming language (lexer parser, tree-walk interpreter, embedding API) in Java with ANTLR4, and shipped production code in Java / Spring Boot during an internship where a single performance fix reduced all API response times by 91 %. Comfortable across Java, Python, C, Rust and Go, with hands on experience with Linux, Docker, MySQL, PostgreSQL, RabbitMQ and Redis. Eligible to work in Ireland and the UK.

Education

University of Limerick

Sept 2022 – Jan 2027

BSc Computer Systems 3.11 QCA Second Class Honours

Professional Experience

Ostia Solutions

Bray, Co. Wicklow

Software Development Intern

May 2024 — Jan 2025

- Worked across test development and performance testing during an 8 month placement, contributing to the quality and reliability of production of Spring Boot Java services.
- Designed and implemented automated test suites using JUnit, JMeter and Postman increasing test coverage and catching bugs earlier in the release cycle.
- Identified and fixed a cache misconfiguration across all major API endpoints, reducing average response time by 91 % and improved throughput under load.
- Investigated and reported performance bottle necks using load testing and profiling, while working alongside senior developers to ship fixes.
- Operated in an agile team environment using git for version control and standard CI work flows for build, test and deployment.

Projects

RJScript — Final Year Project (Java, ANTLR4)

2025 - 2026

Github

- Designed and implemented a dynamically-typed scripting language from scratch in Java (12,000 lines of code), including lexer, parser, AST, and tree-walk interpreter built on ANTLR4.
- Implemented core language features including variables, arithmetic and logical operators with short-circuit evaluation, functions with closures, lexical scoping, strings, floats, arrays with negative indexing, and modulo.

- Built a two-tier standard library and module system, plus a Java embedding API (RJScripEngine) allowing host applications to execute RJScrip code and exchange values with the JVM.
- Added source-location-aware error reporting and configurable debug flags to improve the developer experience when writing and debugging RJScrip programs.

E-Library Mircoservice Backend (Java, Spring Boot, RabbitMQ, Docker)
Github

2026

- Built a microservices-based e-library application split into a core service (catalogue, loans, users) and a support service (fines, notifications), communicating asynchronously via RabbitMQ.
- Used RabbitMQ message queues to decouple fine processing and user notifications from the main request path, improving responsiveness and resilience to downstream failures.
- Containerised all services with Docker and orchestrated local deployment with Docker Compose, including dependency services such as the message broker and database.

Technical Skills

Langauges: Java, Python, C, Rust (basic), Go (basic)

Frameworks & libraries: Spring Boot, ANTLR4, JUnit, Mockito, JMeter

Tools & Platforms: Git, Docker, Docker Compose, Maven, Gradle, RabbitMQ, Linux (Arch)

Databases: MySQL, Redis, PostgreSQL

Concepts: Compilers and interpreters, language design, backend systems, microservices, performance testing, automated testing, message queues, REST APIs